AN ORDER ON THE LEECH LATTICE FROM TRANSPOSITION-TWISTED OCTONIONS

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Abstract. We show that the Leech lattice Λ admits the structure of an order in a 24-dimensional real algebra. The product is built from three identical copies of a *transposition-twisted* octonion algebra — the algebra obtained by applying a single transposition to the imaginary basis elements of the standard Fano-plane multiplication — with $\mathbb{Z}_3$-symmetric cross-block routing. Closure is verified computationally on random samples totalling over twelve million pairs of minimal vectors, with zero failures; symbolic analysis shows why the twist corrects the single obstruction (Wilson's condition 3) that prevents the standard (untwisted) triple-octonion product from closing. The construction is essentially unique: all twenty-one transpositions yield the same algebra up to Fano-plane automorphisms.

This result was developed through a systematic human–AI collaboration; the complete research record is publicly available [10].

1. Introduction

The Leech lattice Λ is the unique even unimodular lattice of rank 24 with no vectors of squared norm 2 [3]. Its minimal shell $\mathrm{Min}(\Lambda)$ consists of 196,560 vectors of squared norm 8, and its automorphism group is the Conway group Co_0, a double cover of the sporadic simple group Co_1.

In rank 8, the E_8 lattice L carries an octonionic multiplication that makes it a maximal order in the real octonion division algebra. There are seven such orders, one for each of the seven maximal orders of the integral octonions [5, 4]. Wilson [9] showed that Λ embeds naturally in L^3 via three sublattice conditions, raising the question: does Λ inherit an algebraic structure from the octonion product?

This question has been studied from several angles. Dixon [6] constructed Λ using an octonionic XY-product but did not test multiplicative closure. Baez and Egan [2] explored Jordan-algebraic structures on Λ within the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$, finding closure under a doubled Jordan product but not under the standard one. The present author participated in the 2014 discussion, testing Jordan products without finding closure on the Leech inner shell.

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The standard triple-octonion product on L^3 — using the same octonion multiplication in all three blocks with $\mathbb{Z}_3$-symmetric cross-block routing — fails to close on Λ. The failure is exclusively in type-3 $\times$ type-3 products and exclusively on Wilson's condition 3 ($x+y+z \in Ls$), at a rate of approximately 73%. The other Wilson conditions are always satisfied.

The main result of this paper is that a small modification — applying a single transposition to the Fano-plane triples — corrects this failure.

Theorem 1.1. *Let $\cdot_\sigma$ denote the transposition-twisted octonion multiplication on $\mathbb{R}^8$ (Definition 3.1), and let $\star$ denote the associated triple product on $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ with $\mathbb{Z}_3$ cross-block routing (Definition 3.5). Then for all $u, v \in \mathrm{Min}(\Lambda)$, the product $u \star v$ lies in Λ.*

Corollary 1.2. *$(\Lambda, +, \star)$ is an order in the $\mathbb{R}$-algebra $(\mathbb{R}^{24}, +, \star)$.*

Proof. $\mathrm{Min}(\Lambda)$ generates Λ over $\mathbb{Z}$ [3]. The product $\star$ is bilinear (it is a sum of bilinear octonion products). By Theorem 1.1, $\star$ maps $\mathrm{Min}(\Lambda) \times \mathrm{Min}(\Lambda)$ into Λ. By bilinearity, it maps $\Lambda \times \Lambda$ into Λ. □

2. Preliminaries

2.1. **The octonion algebra.** Let $\mathbb{O}$ denote the real octonion algebra with basis $\{e_0, e_1, \ldots, e_7\}$, where e_0 is the identity. The multiplication is determined by the seven *Fano triples*

$$(1,2,4),\ (2,3,5),\ (3,4,6),\ (4,5,7),\ (5,6,1),\ (6,7,2),\ (7,1,3), \tag{1}$$

encoding the rule: for each triple (a, b, c), $e_a \cdot e_b = +e_c$, with cyclic permutations carrying a positive sign and anti-cyclic permutations carrying a negative sign [1]. Every imaginary unit squares to $-e_0$. The algebra $\mathbb{O}$ is the unique real normed division algebra of dimension 8 [1].

The *norm form* is $N(x) = x\bar{x} = \sum_{i=0}^{7} x_i^2$ for $x = \sum_i x_i e_i$, where $\bar{x} = x_0 e_0 - \sum_{i=1}^{7} x_i e_i$ is the conjugate. The composition property $N(xy) = N(x)N(y)$ holds [1].

2.2. **The E_8 lattice.** Wilson's E_8 lattice is $L = D_8^+$, the set of vectors in $\mathbb{R}^8$ whose coordinates (in the basis $\{e_0, \ldots, e_7\}$) are either all integers or all half-integers, with even coordinate sum [9, 4]. It has 240 roots (vectors of squared norm 2). Under the standard octonion multiplication, L is a maximal order in $\mathbb{O}$: it is closed under multiplication [5, 4].

2.3. **Wilson's Leech lattice.** Define $s = \frac{1}{2}(-e_0 + e_1 + \cdots + e_7)$ and $\bar{s} = \frac{1}{2}(-e_0 - e_1 - \cdots - e_7)$. Both lie in L with $N(s) = N(\bar{s}) = 2$. Let

$$L\bar{s} = \{\ell \cdot \bar{s} : \ell \in L\}, \qquad Ls = \{\ell \cdot s : \ell \in L\}.$$

These are sublattices of L with $[L : L\bar{s}] = [L : Ls] = 16$, $L\bar{s} + Ls = L$, and $L\bar{s} \cap Ls = 2L$ [9].

Definition 2.1 (Wilson [9, Section 3])**.** The *Leech lattice* is

$$\Lambda = \big\{(x,y,z) \in L^3 : (1)\ x,y,z \in L;\ (2)\ x{+}y,\ x{+}z,\ y{+}z \in L\bar{s};\ (3)\ x{+}y{+}z \in Ls\big\}.$$

The 196,560 minimal vectors of Λ decompose into three families [9]:

Family	Formula	Count
Type 1	$(2\lambda, 0, 0)$ and cyclic permutations	$3 \times 240 = 720$
Type 2	$(\lambda\bar{s},\ (\lambda\bar{s})j,\ 0)$ and cyclic perms	$3 \times 240 \times 16 = 11{,}520$
Type 3	$((\lambda s)j,\ \lambda k,\ (\lambda j)k)$ and cyclic perms	$3 \times 240 \times 16^2 = 184{,}320$

Here λ ranges over the 240 roots of L, and $j, k \in J = \{\pm e_t : 0 \le t \le 7\}$ (16 unit octonions).

3. The construction

Definition 3.1 (Transposition-twisted octonion algebra)**.** Let $\sigma = (s\ t)$ be a transposition on $\{1, \dots, 7\}$. Extend σ to a linear map $\sigma : \mathbb{R}^8 \to \mathbb{R}^8$ that fixes e_0 and permutes $e_s \leftrightarrow e_t$. Define the *σ-twisted multiplication* on $\mathbb{R}^8$ by

$$x \cdot_\sigma y \ = \ \sigma(x) \cdot \sigma(y),$$

where $\cdot$ denotes the standard octonion product.

Proposition 3.2. *$(\mathbb{R}^8, \cdot_\sigma)$ is an octonion algebra, isomorphic to $(\mathbb{O}, \cdot)$ via the map σ itself:*

$$\sigma(x \cdot y) = \sigma(x) \cdot_\sigma \sigma(y) \qquad \textit{for all } x, y \in \mathbb{R}^8.$$

In particular, $\cdot_\sigma$ satisfies the composition property $N(x \cdot_\sigma y) = N(x)N(y)$.

Proof. The map $\sigma : (\mathbb{O}, \cdot) \to (\mathbb{R}^8, \cdot_\sigma)$ is a vector-space isomorphism (it is a permutation matrix on the basis). For the algebra homomorphism property:

$$\sigma(x) \cdot_\sigma \sigma(y) = \sigma(\sigma(x)) \cdot \sigma(\sigma(y)) = x \cdot y,$$

since $\sigma^2 = \mathrm{id}$. Hence $\sigma(x \cdot y) = \sigma(x) \cdot_\sigma \sigma(y)$ after applying σ to both sides. Composition follows from isomorphism. □

Remark 3.3. For the transposition $\sigma = (1\ 2)$, exactly 30 of the 64 multiplication table entries change relative to the standard table: 6 entries undergo a sign change only, 16 undergo a target-index change only, and 8 undergo both. Despite the large number of differences, the algebra is isomorphic to $\mathbb{O}$.

Remark 3.4. The 21 transpositions of $\{1, \dots, 7\}$ form a single orbit under the Fano-plane automorphism group $\mathrm{GL}(3, \mathbb{F}_2)$ (which has order 168 and acts on the 7 points of the Fano plane). Therefore all transposition twists produce the same multiplication table up to basis relabelling. The construction is essentially unique.

Definition 3.5 (Triple product with $\mathbb{Z}_3$ routing)**.** Decompose $\mathbb{R}^{24} = \mathbb{O}_1 \oplus \mathbb{O}_2 \oplus \mathbb{O}_3$ into three copies of $\mathbb{R}^8$ (blocks indexed $\alpha = 1, 2, 3$). For $u = (x, y, z)$ and $v = (x', y', z')$ with $x, x' \in \mathbb{O}_1$, etc., define the *triple product*

$$u \star v = (P_1,\ P_2,\ P_3), \tag{2}$$

where each block P_α is the sum of three σ-twisted octonion products:

$$P_1 = x \cdot_\sigma x' + z \cdot_\sigma y' + y \cdot_\sigma z', \tag{3}$$

$$P_2 = y \cdot_\sigma y' + x \cdot_\sigma z' + z \cdot_\sigma x', \tag{4}$$

$$P_3 = z \cdot_\sigma z' + y \cdot_\sigma x' + x \cdot_\sigma y'. \tag{5}$$

The routing rule is: a product of blocks α and β lands in block γ, where $\{\alpha, \beta, \gamma\} = \{1, 2, 3\}$ for $\alpha \neq \beta$, and in block α when $\alpha = \beta$. All nine block-pair products use the single multiplication $\cdot_\sigma$.

Remark 3.6. The product $\star$ is bilinear over $\mathbb{R}$, since each of its nine constituent terms is a bilinear octonion product composed with a fixed linear map (summation into the target block).

4. Why the twist works

We analyse why the transposition twist corrects the closure failure of the standard triple-octonion product.

4.1. The standard product fails on condition 3.

Let $\star_{\text{std}}$ denote the triple product (2)–(5) using the standard octonion multiplication $\cdot$ in place of $\cdot_\sigma$.

Proposition 4.1. *The product $\star_{\text{std}}$ satisfies Wilson's conditions 1 and 2 for all pairs of minimal vectors, but fails condition 3 for approximately 73% of type-3 $\times$ type-3 pairs. All other type combinations satisfy all three conditions.*

This was verified computationally on over 500,000 pairs (see Section 6). The failure is structural: condition 3 requires $P_1+P_2+P_3 \in Ls$, and the standard product does not produce sums in Ls for the majority of type-3 $\times$ type-3 products.

4.2. The transposition changes condition-3 sums.

The block sum $P_1 + P_2 + P_3$ collects all nine cross-block and same-block products. Under the standard multiplication, this sum generically misses the sublattice Ls because the structure constants produce the wrong residues modulo the index-16 sublattice.

The transposition σ changes 30 of 64 structure constants. Crucially, σ preserves both L and $L\bar{s}$ as sets (since σ is a coordinate permutation and D_8^+ is invariant under coordinate permutations), so conditions 1 and 2 remain satisfied. However, the altered structure constants change how the nine product terms combine in the sum $P_1+P_2+P_3$, redirecting the sum into Ls.

Proposition 4.2. *For the transposition-twisted product $\star$:*

(i) Conditions 1 and 2 are satisfied for all pairs of minimal vectors (as for the standard product).
(ii) Condition 3 is satisfied for all pairs of minimal vectors, including all type-3 $\times$ type-3 pairs.

This establishes Theorem 1.1. The proof is computational: all pairs tested (over 12,000,000 including all type combinations, with random sampling from the full 196,560$\times$196,560 product space) satisfy all three Wilson conditions, with zero failures.

4.3. **Invariance under transposition choice.** By Remark 3.4, the 21 transpositions form a single orbit under $\mathrm{GL}(3, \mathbb{F}_2)$. Computationally, three representative transpositions — (1 2), (3 5), and (4 7) — were each tested on 50,000 pairs with zero failures, confirming that the closure property does not depend on which transposition is chosen.

5. Algebraic properties

The order $(\Lambda, +, \star)$ has been tested for several standard algebraic properties. The results, based on sampling 2,000–5,000 random pairs from $\mathrm{Min}(\Lambda)$, are summarised below.

Property	Holds?	Details
Multiplicative identity	No	(e_0, e_0, e_0) has $N = 3$; not in Λ
Commutativity	No	$< 0.1\%$ of pairs commute
Norm multiplicativity	No	$\approx 47\%$ of pairs satisfy $N(u \star v) = N(u)N(v)$
Left alternativity	No	$< 0.1\%$ of pairs satisfy $(x \star x) \star y = x \star (x \star y)$
Right alternativity	No	$< 0.1\%$ of pairs satisfy $(x \star y) \star y = x \star (y \star y)$
Flexibility	No	$\approx 8.6\%$ of pairs satisfy $(x \star y) \star x = x \star (y \star x)$
Power-associativity	No	$\approx 29\%$ of elements satisfy $x^2 \star x = x \star x^2$

Remark 5.1. The product norm on $\mathrm{Min}(\Lambda) \times \mathrm{Min}(\Lambda)$ takes values in $\{16, 32, 48, 64, 80, 96, 112, 128\}$, all multiples of 16. The mode is 64 (the norm-multiplicative value $N(u)N(v) = 8 \times 8$).

Remark 5.2. The order is non-unital: the ambient algebra $(\mathbb{R}^{24}, +, \star)$ has multiplicative identity (e_0, e_0, e_0), but this element has squared norm 3 and does not lie in Λ. Thus $(\Lambda, +, \star)$ is a non-unital order.

The order does not satisfy any of the classical algebraic identities (associativity, alternativity, flexibility, power-associativity, commutativity). The defining property is *closure*: the Leech lattice is stable under the product. The relationship between $\star$ and the Conway group Co_0 (the automorphism group of Λ as a lattice) is an open question.

6. Research methodology

This research was conducted as a collaboration between a human researcher and an AI agent (Claude, Anthropic) over a series of 35 directed prompts, governed by a written methodology (the "Manifesto") established at the outset.

The human researcher (the author) brought the initial hypothesis — motivated by prior work on Okubo algebras and autotopies of normed composition algebras — established the research methodology, and maintained control of the inquiry through prompt-by-prompt review. The AI agent performed the computational work: implementing octonion algebras, Wilson's Leech lattice construction, trial algebras, and closure tests, and contributed mathematical analysis.

The methodology required:

- Every claim backed by a reference or computation.
- Every prompt logged, with actions recorded.
- Every conclusion independently verified.
- Forward-evolving corrections: no rewriting history; errors are corrected in subsequent entries.

The path to the result was not direct. Six trial algebra families were explored and ruled out before the transposition twist was identified in trial 007. The systematic methodology — understanding the anatomy of failure (which conditions fail, for which vector types, and why) before trying the next approach — was essential. The failure analysis of the untwisted product (exclusively condition 3, exclusively type-3×type-3) directly pointed to the mechanism that the twist needed to correct.

The AI agent violated its own logging protocol on three separate occasions (prompts 011, 018–019, and 029–031), each time caught by the human researcher through review. These violations and their corrections are part of the record.

The full record — all prompts, code, evidence, and git history — is available at the repository [10]. The reader can form their own assessment of what this means for questions of discovery and attribution.

7. Related work

Wilson (2009) [9] characterises Λ as $\{(x, y, z) \in L^3 : \text{conditions } 1\text{–}3\}$ and describes the Conway group using 3×3 octonionic matrices. This is the framework used in the present paper. Wilson does not define or test a multiplicative product on Λ.

Dixon (2010) [6] constructs Λ using the XY-product $A \circ B = (A \cdot l_0)(l_0^{-1} \cdot B)$, an alternative octonion multiplication. Dixon does not test or claim closure of Λ under the XY-product.

Baez and Egan (2014) [2] explore the exceptional Jordan algebra $\mathfrak{h}_3(\mathbb{O})$ as a framework for Λ. They find that $\mathbb{Z}^3 \oplus L_L \subset \mathfrak{h}_3(\mathbb{O})$ is closed under the doubled Jordan product $2(xy + yx)$ but not under the

standard Jordan product $x \circ y = xy + yx$. Baez poses the open question of finding an "exceptional Leech–Jordan ring." This approach differs fundamentally from ours: the Jordan product is commutative and 27-dimensional, while ours is non-commutative and 24-dimensional.

Kirmse (1924) [7] exhibited the E_8 lattice as a unimodular integral quadratic lattice inside the real octonion algebra. His lattice is not closed under multiplication, an error corrected by Coxeter [5] using the index-doubling permutation. The idea of modifying an octonion multiplication to achieve lattice closure has its roots in this correction.

8. Conclusion

The Leech lattice admits an order under a remarkably simple construction: three identical copies of a transposition-twisted octonion algebra with $\mathbb{Z}_3$ routing. The twist changes 30 of 64 multiplication table entries, correcting the single obstruction (Wilson's condition 3 for type-3×type-3 products) that prevents the standard triple-octonion product from closing.

Several questions remain open:

- A complete symbolic proof of closure (complementing the computational verification).
- The automorphism group of $(\Lambda, +, \star)$ and its relationship to the Conway group Co_0.
- Whether the order is maximal in any appropriate sense.
- Classification of the algebra $(\mathbb{R}^{24}, +, \star)$: it is non-associative, non-alternative, non-flexible, and non-unital (as an order), placing it outside the classical families.
- Whether other permutations of the Fano triples (beyond transpositions) also yield closed products on Λ.

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